Supplementary Materials

Electronegativity and chemical hardness of the elements under pressure

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I. Supplementary figures

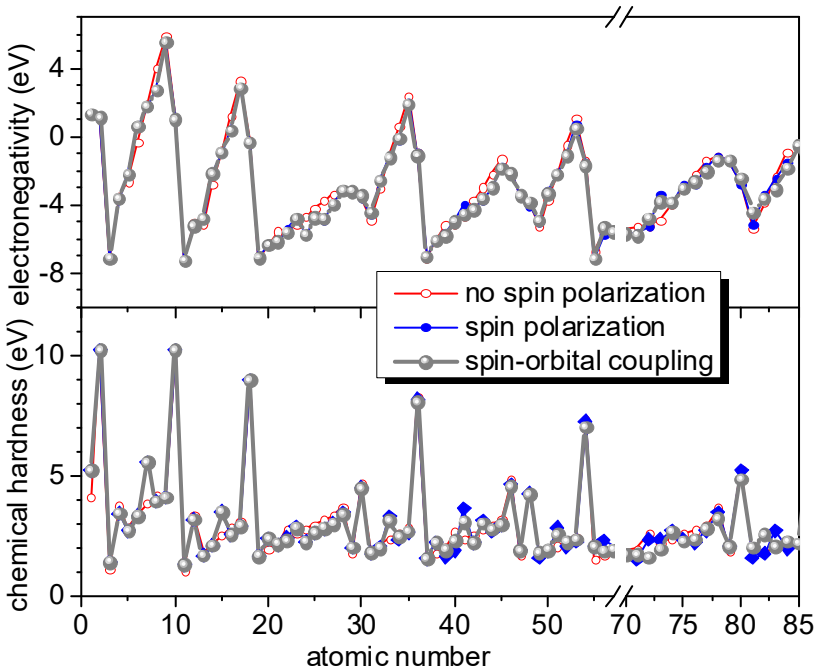

Fig. S1. Electronegativity and chemical hardness at 50 GPa without spin polarization (red), with spin polarization (blue) and with both spin polarization and spin-orbital coupling (grey).

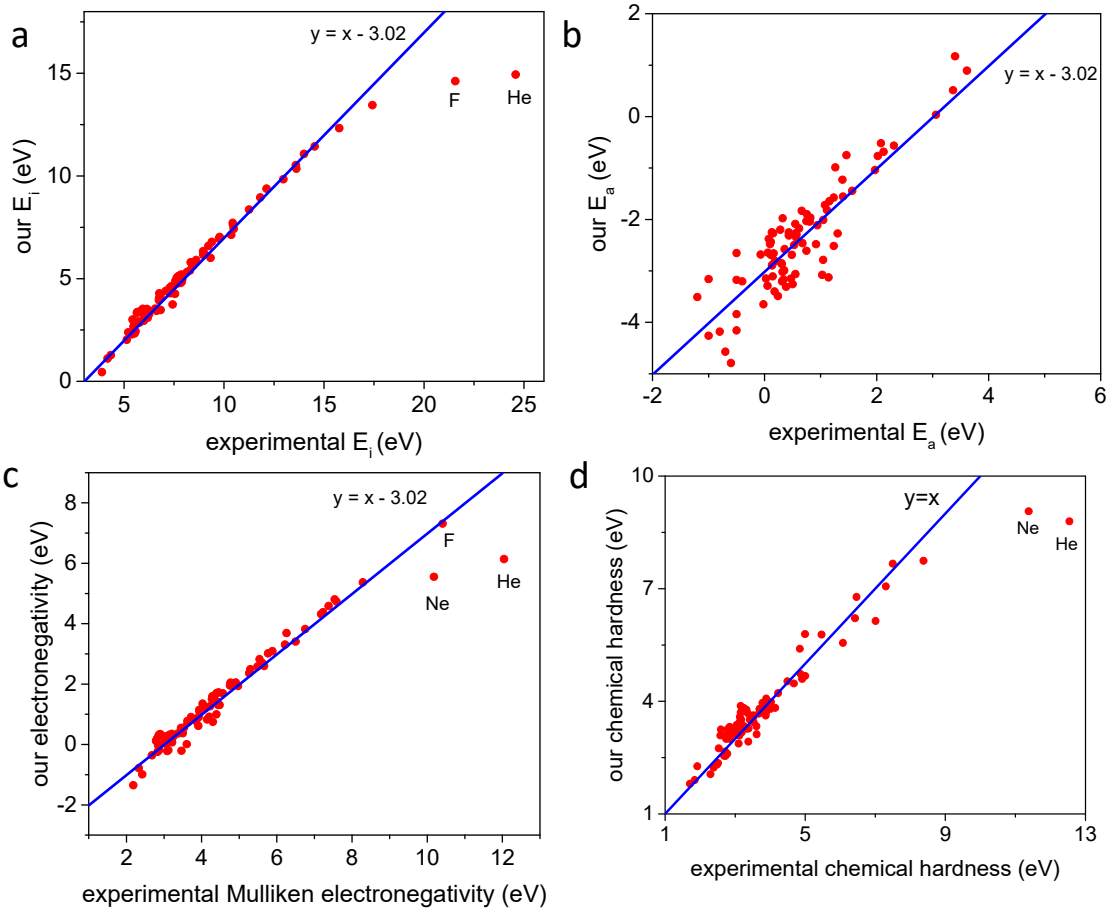

Fig. S2. Comparison of results obtained using our method and experimental data at zero pressure. Correlation between our theoretical values with experimental ones: (a) for the ionization potential (E_i), (b) for the electron affinity (E_a), (c) for the electronegativity, (d) for the chemical hardness. Line in (a-c) shows the best correlation line, which indicates that our electronegativities are related to experimental Mulliken electronegativity by a shift by -3.02 eV (of which -2.11 eV comes from different references - electron gas in our definition, and vacuum in Mulliken's definition; the remaining difference of approximately -0.91 eV is due to approximate DFT functional and effects of He matrix); the same applies to Ei and Ea. Note that here we define ionization potential as the enthalpy of removing an electron from an atom into the electron gas, and electron affinity as minus the enthalpy of taking an electron from electron gas and putting it to the atom.

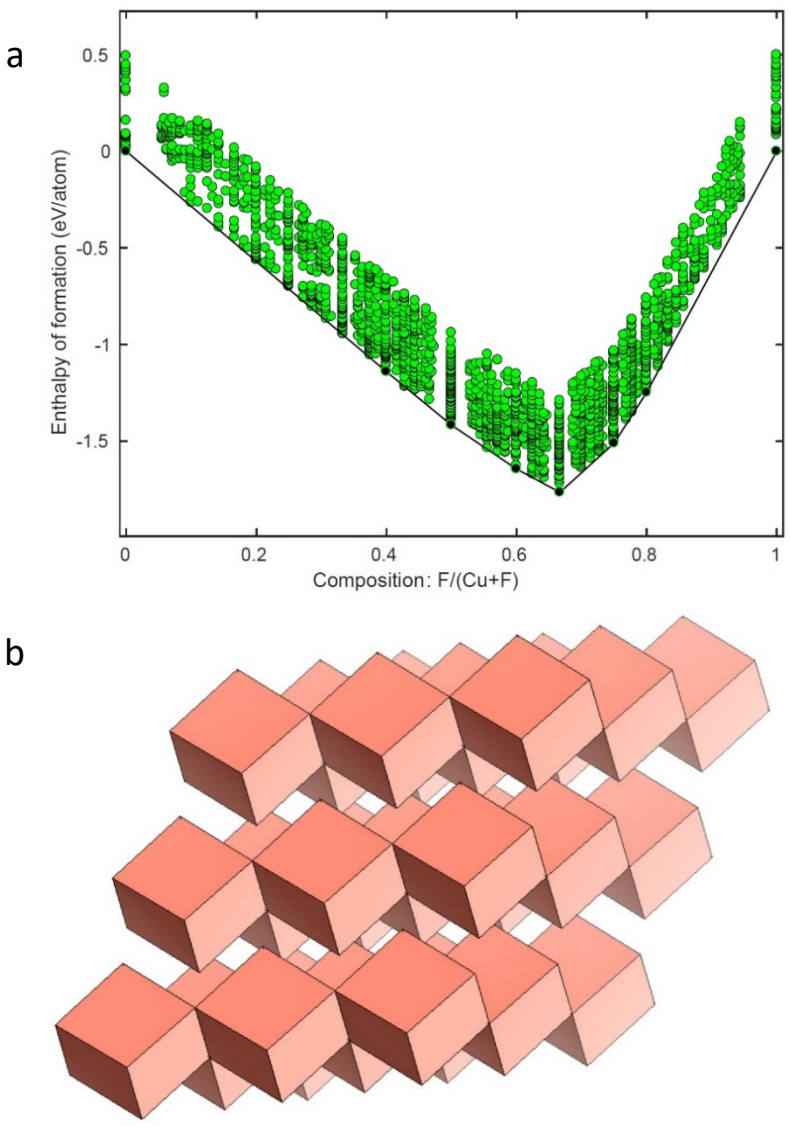

Fig. S3. Cu-F system at 500 GPa: (a) convex hull diagram, showing stability of CuF_4, a compound with an anomalously high valence state of copper, Cu(IV); (b) polyhedral representation of the crystal structure of CuF_4.

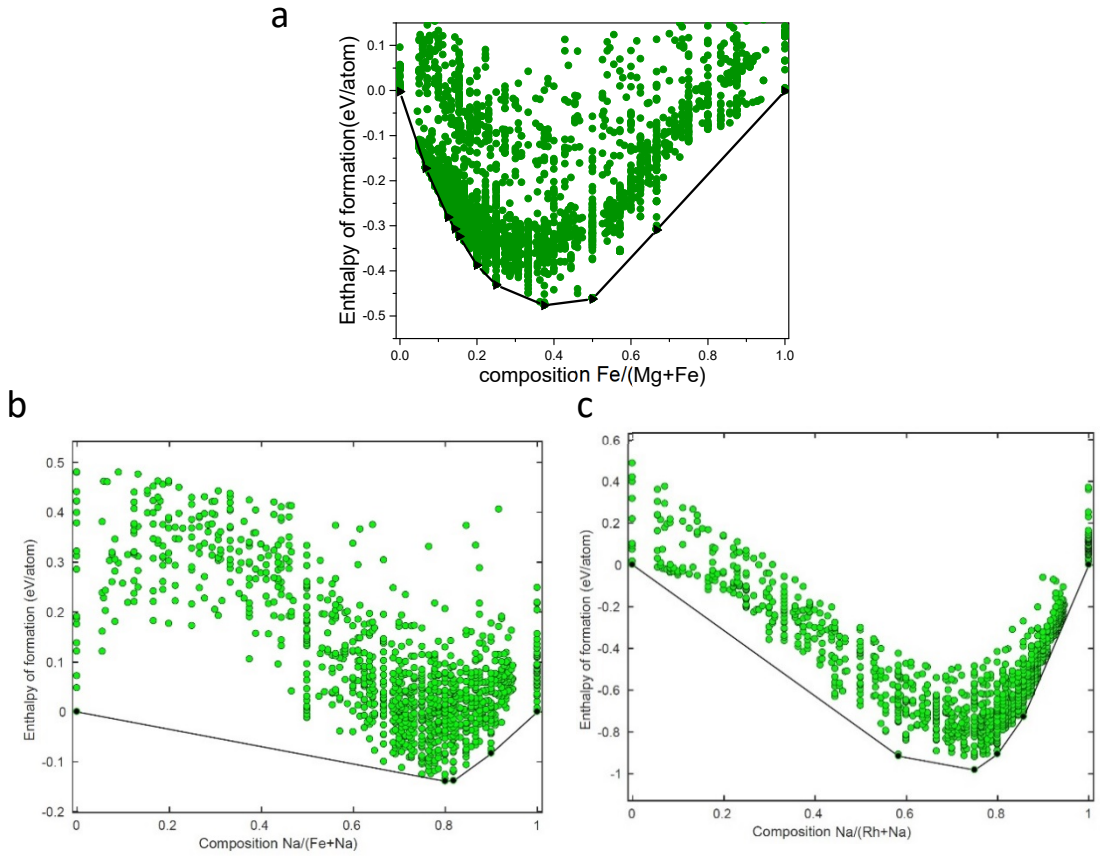

Fig. S4. Thermodynamic convex hulls of systems (a) Mg-Fe, (b) Fe-Na, (c) Rh-Na at 200 GPa. In these systems, there are no stable compounds at normal conditions, but numerous stable compounds appear at high pressure. For example, the list of stable compounds in the Mg-Fe system ($Mg_{14}Fe$, Mg_7Fe, Mg_6Fe, $Mg_{11}Fe_2$, Mg_4Fe, Mg_3Fe, Mg_5Fe_3, MgFe, $MgFe_2$) is very rich, and more complex than previously reported[1].

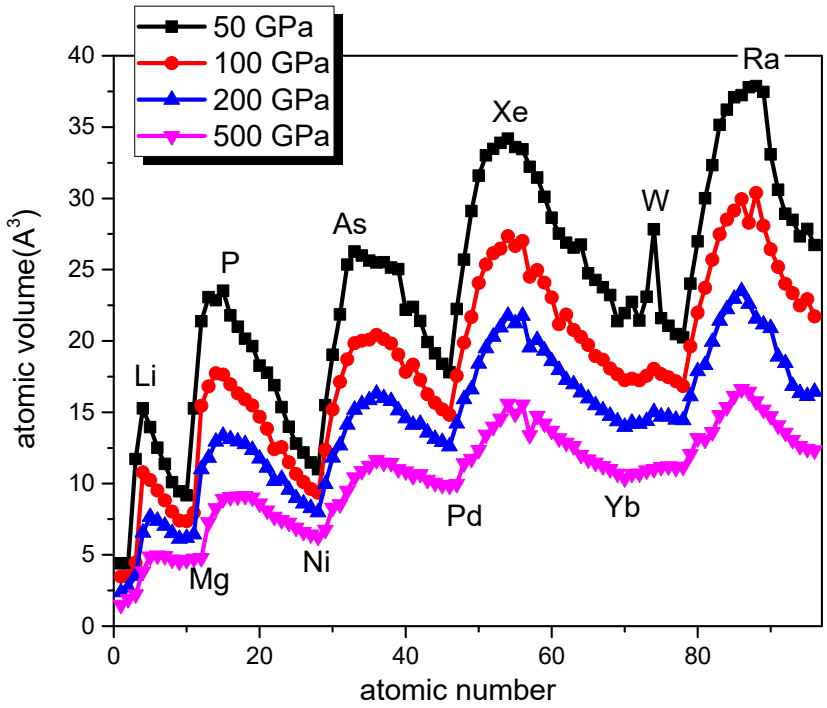

Fig. S5. Non-bonded atomic volumes of the elements at 50, 100, 200, 500 GPa. Note that in many cases minima and maxima of non-bonded volumes correspond to filled and half-filled shells. For example, minima: H (at all pressures, half-filled s-shell), Ne (50 and 200 GPa, filled sp-shell), Ni and Pd (at all pressures, filled sd-shell), Pt (50 and 200 GPa, filled sd-shell). Maxima: Be (50 GPa, filled s-shell), P (50 and 200 GPa, half-filled p-shell), As (50 GPa, half-filled p-shell), Ar (500 GPa, filled sp-shell), Kr (200 and 500 GPa, filled sp-shell), Xe (at all pressures, filled sp-shell), W (50, 200 GPa, half-filled d-shell).

These trends can be easily understood. As pairing makes orbitals more compact, half-filled shells with maximally unpaired electrons correspond to maxima of non-bonded volumes. For the same reason filled shells often correspond to minima of non-bonded volumes, but for heavy noble gases under pressure the situation is opposite – for example, non-bonded volume of Xe is larger than that of Cs, because under pressure the additional electron goes into a compact vacant inner 5d-orbital.

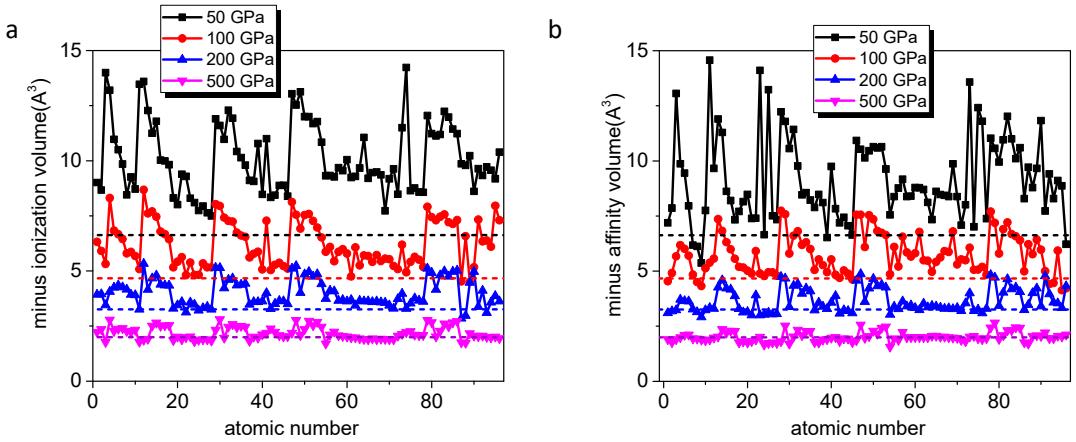

Fig. S6. Minus ionization volume (a) and affinity volume (b) of elements at 50, 100, 200 and 500 GPa. Volumes of the electron in the electron gas at pressures of 50, 100, 200 and 500 GPa are shown by dashed lines.

II. Data on electronegativity and chemical hardness

Table S1. Electronegativities (in eV) of the elements H-Cm at 0, 50, 200, 500 GPa. Note that our electronegativities differ from traditional Mulliken electronegativities due to difference reference systems (electron in the vacuum in Mulliken's electronegativity; electron in electron gas in our electronegativity).

element	0 GPa	50 GPa	200 GPa	500 GPa	element	0 GPa	50 GPa	200 GPa	500 GPa
H	4.32	1.35	0.67	0.36	**In**	-0.22	-4.93	-7.11	-7.82
He	6.14	1.14	0.49	0.20	**Sn**	1.27	-3.35	-5.72	-7.11
Li	0.06	-7.14	-8.06	-7.88	**Sb**	1.95	-2.24	-4.48	-6.04
Be	1.42	-3.66	-6.03	-6.71	**Te**	2.58	-1.10	-3.10	-4.68
B	1.60	-2.20	-3.77	-4.60	**I**	3.82	0.47	-1.41	-3.03
C	3.69	0.61	-0.70	-1.56	**Xe**	2.60	-1.70	-2.64	-3.09
N	4.38	1.75	0.81	0.23	**Cs**	-1.35	-7.17	-7.67	-7.74
O	4.80	2.75	2.23	1.97	**Ba**	-0.36	-5.36	-6.56	-6.90
F	7.31	5.55	5.23	5.06	**La**	0.08	-5.59	-6.30	-7.52
Ne	5.55	1.03	0.38	0.05	**Ce**	0.02	-5.18	-5.93	-6.24
Na	-0.03	-7.28	-8.51	-8.32	**Pr**	0.11	-5.24	-5.95	-6.22
Mg	0.78	-5.23	-7.34	-8.32	**Nd**	0.18	-4.77	-5.97	-6.21
Al	0.35	-4.83	-7.09	-7.69	**Pm**	0.30	-5.30	-5.98	-6.20
Si	2.05	-2.18	-4.67	-6.00	**Sm**	0.34	-4.91	-6.01	-6.20
P	2.69	-0.95	-2.80	-4.30	**Eu**	0.34	-4.98	-6.07	-6.23
S	3.31	0.33	-1.12	-2.47	**Gd**	0.32	-5.44	-6.17	-6.38
Cl	5.37	2.88	1.45	0.42	**Tb**	0.27	-5.13	-6.23	-6.41
Ar	4.58	-0.34	-1.40	-2.06	**Dy**	0.27	-5.54	-6.28	-6.45
K	-0.99	-7.10	-7.54	-7.38	**Ho**	0.15	-5.59	-6.32	-6.47
Ca	0.04	-6.38	-6.90	-6.86	**Er**	0.08	-5.65	-6.37	-6.50
Sc	0.36	-6.14	-6.79	-6.48	**Tm**	0.01	-5.70	-6.39	-6.47
Ti	0.55	-5.60	-6.15	-6.44	**Yb**	-0.19	-5.71	-6.38	-6.43
V	0.73	-4.82	-5.75	-5.64	**Lu**	-0.24	-5.80	-6.51	-6.42
Cr	0.91	-5.76	-5.94	-5.59	**Hf**	0.37	-4.78	-5.71	-6.23
Mn	-0.21	-4.77	-5.48	-5.18	**Ta**	1.14	-3.83	-5.56	-6.00
Fe	1.35	-4.85	-5.09	-4.84	**W**	1.62	-3.88	-4.75	-5.21
Co	1.50	-3.96	-4.20	-3.75	**Re**	1.07	-3.07	-4.10	-4.37
Ni	1.70	-3.18	-4.54	-4.36	**Os**	1.95	-2.64	-3.54	-3.68
Cu	1.31	-3.20	-5.86	-7.71	**Ir**	2.36	-2.04	-2.50	-2.56
Zn	1.00	-3.47	-5.26	-6.25	**Pt**	2.83	-1.39	-2.91	-3.94
Ga	0.15	-4.45	-6.50	-7.21	**Au**	3.01	-1.40	-3.87	-5.94
Ge	1.69	-2.61	-4.77	-6.07	**Hg**	1.94	-2.44	-4.31	-5.36
As	2.49	-1.23	-3.04	-4.30	**Tl**	0.07	-4.44	-6.40	-7.49
Se	3.09	-0.11	-1.69	-2.91	**Pb**	0.86	-3.64	-5.74	-6.85
Br	4.73	1.90	0.47	-0.74	**Bi**	1.22	-3.09	-5.22	-6.76

Kr	3.40	-1.09	-2.12	-2.72	**Po**	2.05	-1.84	-3.71	-5.49
Rb	-0.79	-7.03	-7.73	-7.69	**At**	2.98	-0.51	-2.31	-3.96
Sr	-0.18	-6.13	-6.87	-7.00	**Rn**	1.86	-2.97	-4.30	-5.10
Y	0.28	-5.80	-6.33	-6.46	**Fr**	-0.86	-6.93	-7.77	-7.32
Zr	0.56	-4.98	-5.97	-6.26	**Ra**	-0.52	-6.04	-7.44	-7.42
Nb	0.79	-4.55	-5.24	-5.66	**Ac**	0.06	-5.64	-7.09	-7.91
Mo	0.90	-4.30	-4.59	-4.47	**Th**	0.49	-5.20	-6.53	-7.25
Tc	0.61	-3.65	-4.04	-4.04	**Pa**	-0.07	-5.19	-6.56	-7.20
Ru	0.92	-3.00	-3.43	-3.49	**U**	0.11	-5.28	-6.44	-7.20
Rh	0.75	-1.87	-2.10	-2.00	**Np**	-0.16	-5.40	-6.70	-6.97
Pd	1.72	-2.15	-3.64	-3.74	**Pu**	-0.49	-5.54	-6.68	-6.89
Ag	1.31	-3.46	-6.16	-8.09	**Am**	-0.35	-5.76	-6.88	-7.17
Cd	0.83	-3.85	-5.79	-6.91	**Cm**	-0.21	-5.62	-6.75	-7.06

Table S2. Chemical hardnesses (in eV) of the elements H-Cm at 0, 50, 200, 500 GPa.

element	0 GPa	50 GPa	200 GPa	500 GPa	element	0 GPa	50 GPa	200 GPa	500 GPa
H	6.21	5.22	4.67	4.15	**In**	3.09	1.81	1.41	1.23
He	8.80	10.20	9.77	9.11	**Sn**	3.09	1.91	1.61	1.28
Li	2.23	1.42	1.21	1.21	**Sb**	3.96	2.55	1.96	1.43
Be	4.59	3.43	1.75	1.37	**Te**	3.62	2.25	1.81	1.46
B	3.80	2.71	2.23	1.84	**I**	3.78	2.34	1.88	1.55
C	4.68	3.33	2.74	2.29	**Xe**	6.78	7.02	6.16	5.21
N	7.06	5.53	4.73	4.02	**Cs**	1.80	2.05	1.25	1.20
O	5.55	3.93	3.23	2.72	**Ba**	2.75	1.86	1.71	1.29
F	6.14	4.09	3.27	2.67	**La**	2.35	1.91	1.98	1.21
Ne	9.06	10.20	9.85	9.26	**Ce**	2.32	1.76	1.55	1.46
Na	2.06	1.29	1.16	1.18	**Pr**	2.55	1.79	1.56	1.47
Mg	3.98	3.18	1.68	1.17	**Nd**	2.66	2.34	1.57	1.48
Al	2.60	1.68	1.36	1.23	**Pm**	2.55	1.96	1.58	1.49
Si	3.28	2.09	1.52	1.35	**Sm**	3.00	1.37	1.59	1.50
P	4.73	3.50	2.68	2.21	**Eu**	3.04	1.36	1.59	1.50
S	3.83	2.55	2.04	1.66	**Gd**	3.21	1.74	1.59	1.49
Cl	4.47	2.92	2.27	1.88	**Tb**	3.17	1.37	1.59	1.50
Ar	7.74	8.95	8.24	7.41	**Dy**	3.26	1.73	1.60	1.49
K	2.27	1.67	1.53	1.35	**Ho**	3.32	1.73	1.60	1.49
Ca	3.19	2.37	2.11	1.78	**Er**	3.29	1.72	1.59	1.49
Sc	3.17	2.17	1.89	1.67	**Tm**	3.09	1.71	1.59	1.49
Ti	2.93	2.31	2.17	1.76	**Yb**	3.46	1.71	1.60	1.50
V	3.23	2.82	2.98	2.51	**Lu**	3.25	1.70	1.59	1.31
Cr	3.37	2.25	2.09	1.93	**Hf**	3.77	1.60	1.94	2.03
Mn	3.95	2.62	3.01	2.90	**Ta**	3.12	1.94	2.01	1.74
Fe	3.62	2.83	2.58	2.34	**W**	3.58	2.66	1.88	1.71
Co	3.33	3.00	2.69	2.53	**Re**	3.72	2.28	2.62	2.24
Ni	3.34	3.40	4.52	4.60	**Os**	3.67	2.35	2.32	2.01
Cu	3.82	2.02	1.57	1.09	**Ir**	3.80	2.79	2.44	2.12
Zn	5.79	4.48	3.42	2.19	**Pt**	3.51	3.23	4.04	4.74
Ga	3.00	1.79	1.37	1.29	**Au**	3.58	2.03	1.59	1.24
Ge	3.27	2.00	1.59	1.28	**Hg**	5.78	4.86	4.16	3.07
As	4.53	3.14	2.52	1.93	**Tl**	3.09	2.02	1.57	1.29
Se	3.86	2.45	1.97	1.60	**Pb**	3.44	2.56	2.29	1.71
Br	4.22	2.68	2.14	1.82	**Bi**	3.33	2.03	1.57	1.35
Kr	7.67	8.06	7.32	6.44	**Po**	3.60	2.28	1.72	1.40
Rb	1.90	1.53	1.31	1.18	**At**	3.56	2.19	1.74	1.43
Sr	3.11	2.21	1.86	1.39	**Rn**	6.05	6.17	5.15	4.02
Y	3.14	1.90	1.61	1.50	**Fr**	1.80	1.33	1.37	1.44

Zr	2.87	2.30	2.03	2.06	**Ra**	3.02	1.92	1.42	1.79
Nb	3.27	3.08	2.45	1.72	**Ac**	2.84	1.81	1.24	1.13
Mo	3.51	2.22	1.96	2.18	**Th**	2.77	1.52	1.46	1.30
Tc	3.68	3.02	2.84	2.68	**Pa**	3.14	2.12	1.39	1.59
Ru	3.70	2.77	2.40	2.07	**U**	3.01	2.22	1.42	1.44
Rh	3.88	2.97	2.59	2.24	**Np**	3.11	2.15	1.42	1.62
Pd	4.07	4.55	5.47	5.39	**Pu**	3.46	1.90	1.44	1.97
Ag	3.58	1.89	1.44	1.10	**Am**	3.28	2.45	1.77	1.71
Cd	5.40	4.20	3.19	1.93	**Cm**	3.32	2.28	1.59	2.14

III. Volumetric data

Table S3. Non-bonded atomic volume (in Å^3) of the elements H-Cm in He matrix at 50, 200, 500 GPa. In a bonded system the atoms adopt sizes at which bonding (attraction) and repulsion balance each other. In a He matrix there is (depending on level of theory) little or no bonding, as He atoms display only repulsion to each other and to other atoms. For this reason, non-bonded volumes at 0 GPa will be very large or infinite, corresponding to infinite size of a free atom. On increasing pressure, these volumes become well-defined and useful, and reflect the size at which the compressed atom balances external pressure. For comparison, the volume occupied by an electron in homogeneous electron gas is 45.498 Å^3/electron at 0 GPa, 6.626 Å^3/electron at 50 GPa, 3.254 Å^3/electron at 200 GPa, 1.989 Å^3/electron at 500 GPa.

element	50 GPa	200 GPa	500 GPa	element	50 GPa	200 GPa	500 GPa
H	4.40	2.38	1.49	**In**	29.11	16.60	11.74
He	4.40	2.71	1.90	**Sn**	31.59	18.40	12.35
Li	11.71	3.53	2.23	**Sb**	33.01	19.49	13.43
Be	15.27	6.56	3.83	**Te**	33.47	20.30	13.99
B	13.95	7.66	4.88	**I**	33.89	20.95	14.55
C	12.50	7.39	4.96	**Xe**	34.18	21.77	15.59
N	11.37	7.04	4.91	**Cs**	33.59	21.23	14.89
O	10.09	6.52	4.67	**Ba**	33.44	21.77	15.54
F	9.45	6.13	4.52	**La**	32.21	19.55	13.39
Ne	9.17	6.19	4.64	**Ce**	31.45	20.07	14.75
Na	15.27	6.43	4.73	**Pr**	30.10	19.32	14.20
Mg	21.38	11.00	4.78	**Nd**	28.63	18.59	13.69
Al	23.07	11.81	7.28	**Pm**	27.52	17.96	13.17
Si	22.85	12.94	8.26	**Sm**	26.89	17.26	12.87
P	23.52	13.35	8.92	**Eu**	26.54	16.96	12.62
S	21.79	13.12	9.05	**Gd**	26.75	16.39	12.01
Cl	20.99	12.95	9.10	**Tb**	24.74	15.97	11.69
Ar	20.16	12.77	9.12	**Dy**	24.28	15.50	11.46
K	19.64	12.36	9.02	**Ho**	23.77	15.13	11.21
Ca	18.27	11.75	8.60	**Er**	23.21	14.75	11.01
Sc	17.77	11.12	8.11	**Tm**	21.38	14.36	10.69
Ti	16.90	10.18	7.67	**Yb**	21.94	13.98	10.37
V	15.36	10.34	7.43	**Lu**	22.74	14.27	10.67
Cr	13.98	9.56	7.25	**Hf**	21.42	14.20	10.71
Mn	12.80	8.98	6.90	**Ta**	23.09	14.39	10.91
Fe	12.16	8.60	6.61	**W**	27.83	15.01	11.01

Co	11.47	8.28	6.43	**Re**	21.59	14.69	11.16
Ni	11.00	7.98	6.27	**Os**	21.06	14.70	11.20
Cu	15.49	9.97	6.75	**Ir**	20.47	14.48	11.10
Zn	19.02	11.82	8.29	**Pt**	20.25	14.44	11.18
Ga	21.85	12.68	8.54	**Au**	24.01	16.12	12.06
Ge	25.35	14.14	9.49	**Hg**	26.98	17.88	13.18
As	26.27	15.18	10.43	**Tl**	30.00	18.32	13.08
Se	25.98	15.56	10.87	**Pb**	32.32	19.95	13.60
Br	25.63	15.85	11.24	**Bi**	35.15	21.42	14.80
Kr	25.50	16.30	11.65	**Po**	36.22	22.23	15.36
Rb	25.52	16.00	11.46	**At**	37.09	22.95	16.15
Sr	25.15	15.79	11.44	**Rn**	37.24	23.50	16.62
Y	25.03	15.13	11.00	**Fr**	37.79	22.61	16.43
Zr	22.18	14.57	10.85	**Ra**	37.88	21.56	15.76
Nb	22.38	14.13	10.51	**Ac**	37.46	21.16	15.18
Mo	21.40	14.19	10.66	**Th**	33.08	20.91	14.72
Tc	19.92	13.57	10.26	**Pa**	30.60	18.90	14.08
Ru	19.13	13.13	10.05	**U**	28.92	18.45	13.57
Rh	18.39	12.92	9.90	**Np**	28.49	16.85	13.06
Pd	17.80	12.61	9.85	**Pu**	27.33	16.35	12.63
Ag	22.24	14.22	9.97	**Am**	27.85	16.14	12.43
Cd	25.70	15.92	11.42	**Cm**	26.72	16.44	12.32

Table S4. Minus ionization volume (in Å^3) of the elements H-Cm in He matrix at 50, 200, 500 GPa. For comparison, the volume occupied by an electron in homogeneous electron gas is 45.498 Å^3/electron at 0 GPa, 6.626 Å^3/electron at 50 GPa, 3.254 Å^3/electron at 200 GPa, 1.989 Å^3/electron at 500 GPa.

element	50 GPa	200 GPa	500 GPa	element	50 GPa	200 GPa	500 GPa
H	9.01	3.93	2.23	**In**	13.13	4.01	2.08
He	8.67	3.95	2.35	**Sn**	12.00	4.85	2.35
Li	14.00	3.45	1.77	**Sb**	12.00	4.96	2.73
Be	13.20	4.06	2.78	**Te**	11.69	4.75	2.53
B	10.97	4.25	2.27	**I**	11.78	4.86	2.68
C	10.50	4.35	2.37	**Xe**	10.84	4.41	2.46
N	9.85	4.26	2.40	**Cs**	9.32	3.74	1.70
O	8.45	4.16	2.27	**Ba**	9.33	4.11	2.11
F	9.26	3.93	2.22	**La**	9.27	4.08	2.24
Ne	8.72	3.91	2.32	**Ce**	9.68	3.65	2.07
Na	13.47	3.25	1.78	**Pr**	9.55	3.69	2.04
Mg	13.60	5.33	1.87	**Nd**	10.05	3.62	2.02

Al	12.28	4.15	1.91	**Pm**	9.24	3.69	1.98
Si	11.25	4.67	2.51	**Sm**	9.29	3.44	1.98
P	11.79	4.80	2.65	**Eu**	9.67	3.65	1.93
S	10.03	4.39	2.50	**Gd**	11.06	3.62	1.90
Cl	9.99	4.35	2.53	**Tb**	9.20	3.63	1.89
Ar	9.82	4.34	2.54	**Dy**	9.46	3.59	1.93
K	8.31	3.30	1.85	**Ho**	9.49	3.62	1.89
Ca	8.01	3.45	2.01	**Er**	9.36	3.58	1.92
Sc	9.39	3.59	1.89	**Tm**	7.73	3.56	1.91
Ti	9.29	3.12	1.99	**Yb**	9.18	3.44	1.89
V	8.30	3.57	2.02	**Lu**	9.63	3.30	1.89
Cr	8.10	3.48	1.80	**Hf**	8.48	3.73	2.06
Mn	7.74	3.21	1.87	**Ta**	11.50	3.97	2.14
Fe	7.94	3.32	1.89	**W**	14.23	3.29	2.22
Co	7.64	3.38	1.89	**Re**	8.64	3.53	2.26
Ni	7.49	3.25	1.83	**Os**	8.76	3.76	2.10
Cu	11.90	5.15	2.32	**Ir**	8.58	3.66	2.08
Zn	11.60	5.14	2.80	**Pt**	8.57	3.61	2.09
Ga	10.97	4.22	1.96	**Au**	12.05	5.11	2.77
Ge	12.29	4.60	2.46	**Hg**	11.23	4.92	2.67
As	11.93	4.67	2.58	**Tl**	11.13	4.15	2.02
Se	10.42	4.35	2.44	**Pb**	11.19	4.81	2.27
Br	10.14	4.38	2.50	**Bi**	12.24	4.99	2.61
Kr	9.80	4.43	2.46	**Po**	11.98	4.75	2.48
Rb	9.11	3.48	1.82	**At**	11.44	4.97	2.66
Sr	9.06	3.59	2.01	**Rn**	11.23	5.02	2.72
Y	10.78	3.59	1.96	**Fr**	9.86	2.85	1.75
Zr	8.47	3.62	2.12	**Ra**	9.79	2.97	1.73
Nb	11.00	4.00	2.19	**Ac**	10.23	4.47	2.17
Mo	8.34	3.28	2.38	**Th**	8.61	4.97	2.06
Tc	8.44	3.51	2.23	**Pa**	9.64	3.38	2.03
Ru	8.88	3.64	2.08	**U**	9.33	4.11	2.06
Rh	8.89	3.63	2.02	**Np**	9.72	3.10	2.00
Pd	8.39	3.50	2.03	**Pu**	9.57	3.57	1.98
Ag	13.04	5.11	2.19	**Am**	9.18	3.87	2.01
Cd	12.53	5.23	2.77	**Cm**	10.39	3.62	1.94

Table S5. Minus affinity volume (in Å^3) of the elements H-Cm in He matrix at 50, 200, 500 GPa. For comparison, the volume occupied by an electron in homogeneous electron gas is 45.498 Å^3/electron at 0 GPa, 6.626 Å^3/electron at 50 GPa, 3.254 Å^3/electron at 200 GPa, 1.989 Å^3/electron at 500 GPa.

element	50 GPa	200 GPa	500 GPa	element	50 GPa	200 GPa	500 GPa
H	7.19	3.10	1.90	**In**	10.46	4.22	2.04
He	7.86	3.18	1.77	**Sn**	10.64	4.65	2.29
Li	13.06	3.23	1.90	**Sb**	10.59	4.32	2.15
Be	9.87	3.71	1.99	**Te**	10.64	4.39	2.42
B	9.45	3.64	2.10	**I**	9.63	4.26	2.48
C	7.96	3.63	2.12	**Xe**	7.55	3.03	1.56
N	6.18	3.27	1.94	**Cs**	8.38	3.39	1.94
O	6.12	3.14	1.93	**Ba**	8.77	3.28	1.87
F	5.37	2.91	1.88	**La**	9.17	3.63	2.23
Ne	7.75	3.23	1.85	**Ce**	8.38	3.41	1.99
Na	14.57	3.33	1.89	**Pr**	8.40	3.45	2.00
Mg	9.66	3.30	1.98	**Nd**	8.79	3.33	2.00
Al	11.90	4.26	2.02	**Pm**	8.78	3.24	2.00
Si	11.29	4.59	2.36	**Sm**	8.70	3.53	1.99
P	8.62	4.22	2.20	**Eu**	8.12	3.36	1.99
S	8.20	4.17	2.29	**Gd**	7.34	3.43	2.06
Cl	7.33	3.87	2.27	**Tb**	8.62	3.40	2.06
Ar	7.71	3.29	1.76	**Dy**	8.47	3.35	2.01
K	8.15	3.13	1.80	**Ho**	8.44	3.30	2.02
Ca	8.48	3.15	1.75	**Er**	8.38	3.32	1.99
Sc	7.39	3.00	1.82	**Tm**	9.87	3.28	1.98
Ti	7.40	3.91	1.89	**Yb**	8.38	3.33	1.96
V	14.11	3.00	2.01	**Lu**	7.09	3.19	1.87
Cr	6.65	3.03	1.66	**Hf**	8.01	3.54	1.81
Mn	13.23	3.11	1.73	**Ta**	13.58	4.00	2.02
Fe	7.51	3.04	1.76	**W**	7.01	3.19	2.05
Co	7.34	3.06	1.73	**Re**	12.42	3.26	1.95
Ni	12.23	4.74	1.81	**Os**	11.80	3.09	1.90
Cu	11.79	4.67	2.52	**Ir**	7.38	3.30	1.95
Zn	10.56	3.38	1.70	**Pt**	11.03	4.83	2.43
Ga	11.43	4.34	2.03	**Au**	10.57	4.71	2.66
Ge	9.77	4.42	2.33	**Hg**	9.95	3.43	1.91
As	8.46	3.98	2.09	**Tl**	10.96	4.03	2.09
Se	8.56	4.22	2.27	**Pb**	12.02	4.64	2.31
Br	8.22	4.09	2.27	**Bi**	11.00	4.18	2.31
Kr	7.89	3.22	1.77	**Po**	10.10	4.26	2.44
Rb	8.48	3.57	1.75	**At**	10.60	4.06	2.42
Sr	8.11	3.39	1.89	**Rn**	8.29	3.24	1.76
Y	6.52	3.29	1.90	**Fr**	9.71	3.47	1.70
Zr	9.75	3.57	1.97	**Ra**	8.79	4.01	2.04

Nb	7.83	3.88	2.00	**Ac**	9.66	4.22	2.08
Mo	7.19	3.08	1.82	**Th**	11.83	3.41	2.05
Tc	7.44	3.23	1.94	**Pa**	7.73	4.78	2.21
Ru	7.06	3.19	1.93	**U**	9.41	3.96	1.95
Rh	6.64	3.09	1.77	**Np**	8.29	3.58	1.95
Pd	10.93	4.57	1.85	**Pu**	9.11	3.54	2.06
Ag	10.53	4.87	2.57	**Am**	8.87	3.33	2.03
Cd	10.14	3.61	1.96	**Cm**	6.22	4.32	2.12

IV. Electronic configurations and magnetic moments of the atoms.

Table S6. Calculated magnetic moments (in μ_B, upper row) valence electronic configurations (lower rows) and of the elements H-Cm in He matrix at 0, 50, 200, 500 GPa.

Element	0 GPa	50 GPa	200 GPa	500 GPa	Element	0 GPa	50 GPa	200 GPa	500 GPa
H	1 s1	1 s1	1 s1	1 s1	**In**	1 s2p1	1 s2p1	0 s2p1	0 s2p1
He	0 s2	0 s2	0 s2	0 s2	**Sn**	2 s2p2	2 s2p2	2 s2p2	0 s2p2
Li	1 s1	1 s1	0 s0p1	0 s0p1	**Sb**	3 s2p3	3 s2p3	3 s2p3	0 s2p3
Be	0.06 s2	0.06 s2	2 s1p1	2 s1p1	**Te**	2 s2p4	2 s2p4	2 s2p4	2 s2p4
B	1 s2p1	1 s2p1	1 s2p1	1 s2p1	**I**	1 s2p5	1 s2p5	1 s2p5	1 s2p5
C	2 s2p2	2 s2p2	2 s2p2	2 s2p2	**Xe**	0.14 s2p6	0.15 s2p6	0.16 s2p6	0 s2p6
N	3 s2p3	3 s2p3	3 s2p3	3 s2p3	**Cs**	0.97 s1	1 d1s0	0 d1s0	0 d1s0
O	2 s2p4	2 s2p4	2 s2p4	2 s2p4	**Ba**	0.06 d0S2	2 d2s0	0 d2s0	0 d2s0
F	1 s2p5	1 s2p5	1 s2p5	1 s2p5	**La**	3 f0d2s1	3 f0d3s0	1 f1d2s0	0 f2d1s0
Ne	0.06 s2p6	0.10 s2p6	0 s22p6	0 s2p6	**Ce**	3 [f1]d2s1	1 [f1]d3s0	1 [f1]d3s0	1 [f1]d3s0
Na	1 s1	1 s1	0 s0p1	0 s0p1	**Pr**	3 [f2]d2s1	1 [f2]d3s0	1 [f2]d3s0	0 [f2]d3s0
Mg	0.07 s2	0.06 s2	0.05 s2	0 s2	**Nd**	3.07 [f3]d2s1	1 [f3]d3s0	1 [f3]d3s0	1 [f3]d3s0
Al	1 s2p1	1 s2p1	0.01 s2p1	0 s2p1	**Pm**	2.89 [f4]d2s1	1 [f4]d3s0	1 [f4]d3s0	1 [f4]d3s0
Si	2 s2p2	2 s2p2	2 s2p2	0 s2p2	**Sm**	1 [f5]d1s2	1 [f5]d3s0	1 [f5]d3s0	1 [f5]d3s0
P	3 s2p3	3 s2p3	3 s2p3	3 s2p3	**Eu**	1 [f6]d1s2	1 [f6]d3s0	1 [f6]d3s0	1 [f6]d3s0
S	2 s2p4	2 s2p4	2 s2p4	2 s2p4	**Gd**	1 [f7]d1s2	1 [f7]d3s0	1 [f7]d3s0	1 [f7]d3s0
Cl	1 s2p5	1 s2p5	1 s2p5	1 s2p5	**Tb**	1 [f8]d1s2	1 [f8]d3s0	1 [f8]d3s0	1 [f8]d3s0
Ar	0.11 s2p6	0.12 s2p6	0.13 s2p6	0.09 s2p6	**Dy**	1 [f9]d1s2	1 [f9]d3s0	1 [f9]d3s0	1 [f9]d3s0
K	1.04 s1	1 d1s0	1 d1s0	0 d1s0	**Ho**	1 [f10]d1s2	1 [f10]d3s0	1 [f10]d3s0	1 [f10]d3s0
Ca	0.07 s2	2 d2s0	2 d2s0	2 d2s0	**Er**	1 [f11]d1s2	1 [f11]d3s0	1 [f11]d3s0	1 [f11]d3s0
Sc	0.99 d1s2	3 d3s0	1 d3s0	1 d3s0	**Tm**	1 [f12]d1s2	1 [f12]d3s0	1 [f12]d3s0	1 [f12]d3s0
Ti	3.41 d3s1	4 d4s0	4 d4s0	0.01 d4s0	**Yb**	1 [f13]d1s2	1 [f13]d3s0	1 [f13]d3s0	1 [f13]d3s0
V	5 d4s1	5 d5s0	5 d5s0	5 d5s0	**Lu**	1 [f14]d1s2	1 [f14]d3s0	1 [f14]d3s0	1 [f14]d3s0

Cr	6 d5s1	4 d6s0	4 d6s0	4 d6s0	**Hf**	2 d2s2	4 d4s0	0 d4s0	0 d4s0
Mn	5 d5s2	3 d7s0	3 d7s0	3 d7s0	**Ta**	3 d3s2	5 d5s0	1 d5s0	1 d5s0
Fe	4 d6s2	2 d8s0	2 d8s0	2 d8s0	**W**	6 d5s1	6 d5s1	1.99 d6s0	2 d6s0
Co	3 d7s2	1 d9s0	1 d9s0	1 d9s0	**Re**	5 d5s2	3 d7s0	3 d7s0	2.99 d7s0
Ni	2 d9s1	0.10 d10s0	0.10 d10s0	0.10 d10s0	**Os**	4 d6s2	2 d8s0	2 d8s0	2 d8s0
Cu	0.89 d10s1	1 d10s1	1 d10s1	0 d10s1	**Ir**	3 d8s1	1 d9s0	1 d9s0	1 d9s0
Zn	0.07 d10s2	0 d10s2	0.07 d10s2	0.05 d10s2	**Pt**	2 d9s1	0.13 d10s0	0.13 d10s0	0.12 d10s0
Ga	1 s2p1	1 s2p1	0 s2p1	0 s2p1	**Au**	1 d10s1	1 d10s1	1 d10s1	0 d10s1
Ge	2 s2p2	2 s2p2	2 s2p2	0 s2p2	**Hg**	0.07 d10s2	0.07 d10s2	0.05 d10s2	0 d10s2
As	3 s2p3	3 s2p3	3 s2p3	3 s2p3	**Tl**	1 s2p1	10 s2p1	0 s2p1	0 s2p1
Se	2 s2p4	2 s2p4	2 s2p4	2 s2p4	**Pb**	2 s2p2	2 s2p2	2 s2p2	0 s2p2
Br	1 s2p5	1 s2p5	1 s2p5	1 s2p5	**Bi**	3 s2p3	3 s2p3	3 s2p3	0 s2p3
Kr	0.12 s2p6	0.13 s2p6	0.14 s2p6	0.13 s2p6	**Po**	2 s2p4	2 s2p4	2 s2p4	2 s2p4
Rb	1.02 s1	1 d1s0	0 d1s0	0 d1s0	**At**	1 s2p5	1 s2p5	1 s2p5	1 s2p5
Sr	0.07 s2	2 d2s0	2 d2s0	0 d2s0	**Rn**	0 s2p6	0.15 s2p6	0 s2p6	0 s2p6
Y	1 d1s2	1 d3s0	1 d3s0	1 d3s0	**Fr**	1 d0s1	1 d1s0	0 d1s0	0 d1s0
Zr	3.94 d3s1	4 d4s0	0.03 d4s0	0 d4s0	**Ra**	0 f0d0s2	0 f0d2s0	0 f0d2s0	0 f1d1s0
Nb	5 d4s1	5 d5s0	1 d5s0	1 d5s0	**Ac**	1 f0d1s2	3 f0d3s0	1 f1d2s0	0 f2d1s0
Mo	6 d5s1	4 d6s0	4 d6s0	2 d6s0	**Th**	2 f0d2s2	4 f2d2s0	2 f2d2s0	2 f2d2s0
Tc	5 d5s2	3 d7s0	3 d7s0	3.06 d7s0	**Pa**	3 f2d1s2	5 f3d2s0	3 f4d1s0	3 f4d1s0
Ru	4 d6s2	2 d8s0	2 d8s0	2 d8s0	**U**	4 f4d0s2	6 f4d2s0	4 f5d1s0	2.12 f6d0s0
Rh	1 d9s0	1 d9s0	1 d9s0	1 d9s0	**Np**	5 f5d0s2	7 f5d2s0	1 f7d0s0	1 f7d0s0
Pd	0.14 d10s0	0.12 d10s0	0.12 d10s0	0.12 d10s0	**Pu**	6 f6d0s2	8 f6d2s0	0 f8d0s0	0.16 f8d0s0
Ag	1 d10s1	1 d10s1	0 d10s1	0 d10s1	**Am**	7 f7d0s2	9 f7d2s0	1 f9d0s0	1 f9d0s0
Cd	0.07 d10s2	0.07 d10s2	0.06 d10s2	0 d10s2	**Cm**	6.72 f8d0s2	8 f8d2s0	4.01 f10d0s0	2 f10d0s0

The main conclusions are:

(1) At normal conditions, for most elements DFT predicts correct ground states. Even for those elements where this is not the case (for some transition metals – e.g. DFT predicts d^3s^1 for Ti, whereas the true ground state is d^2s^2), DFT still correctly predicts chemistry and chemical properties of the element.

(2) Many elements change their electronic configuration under pressure, although a large number of sp-elements keep the same electronic configuration throughout studied pressure range – e.g., B, C, N, O, F, P, S, Cl, etc. Transition metals from 5-10 groups typically display electronic transitions (often already at pressures <50 GPa). While most transition elements show just one electronic transition, some heavy transition metals show two (Ta, W, Re) in the studied pressure range.
(3) Under pressure we observe transitions of the electrons to available orbitals with higher angular momenta (e.g., electrons go from s- to p- or d-orbitals under pressure);
(4) Overall, magnetic moments decrease under pressure – one notable exception is electronic transitions where electrons go from a filled s^2-subshell to p- or d-orbitals and the number of unpaired electrons increases. For example, alkali earth metals undergo s-p or s-d transitions (Be at <200 GPa, while Ca, Sr, Ba at pressures below 50 GPa), which causes electron unpairing – as a result, these nonmagnetic atoms become magnetic under pressure. At higher pressures, pairing happens again – in Sr and Ba at pressures >200 GPa, when magnetic moment becomes zero again. Group-3 elements initially have magnetic moment $1\mu_B$, which then increases to $3\mu_B$ (at <50 GPa), and then (at > 200 GPa for Sc and <200 GPa for Y and Lu) decreases back to $1\mu_B$.
(5) At very high pressures magnetic moments are completely suppressed, even for atoms with an odd number of electrons. This can happen when these electrons form electronic bands with the matrix (this happens even in the most inert He matrix) – but for this to happen, the element should have very low electronegativity. This happens, for example, at pressures below 200 GPa for Li, Na, Rb and Cs and at >200 GPa for K); Al, Ga, In, Tl (at ~200 GPa or above), Cu and Ag (at >200 GPa). We also see many cases when at high pressure such configurations as d^2, d^4, etc. become non-magnetic.

References

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